Supplemental Material for "Learning impurity incorporation energetics across host and impurity chemistry in two-dimensional materials with a defect-aware graph Transformer"

Yiming Hua[1]

[1] *Wuhan University, Wuhan, China*

(Dated: August 13, 2026)

I. DATA AUDIT AND FIXED EVALUATION PARTITIONS

The retained modeling set is obtained through a sequential source audit, target-provenance check, unique-impurity requirement, and duplicate-control procedure. Figure S1 summarizes those exclusions, the formation-energy distribution, chemical coverage, and the fixed evaluation partitions used throughout the study.

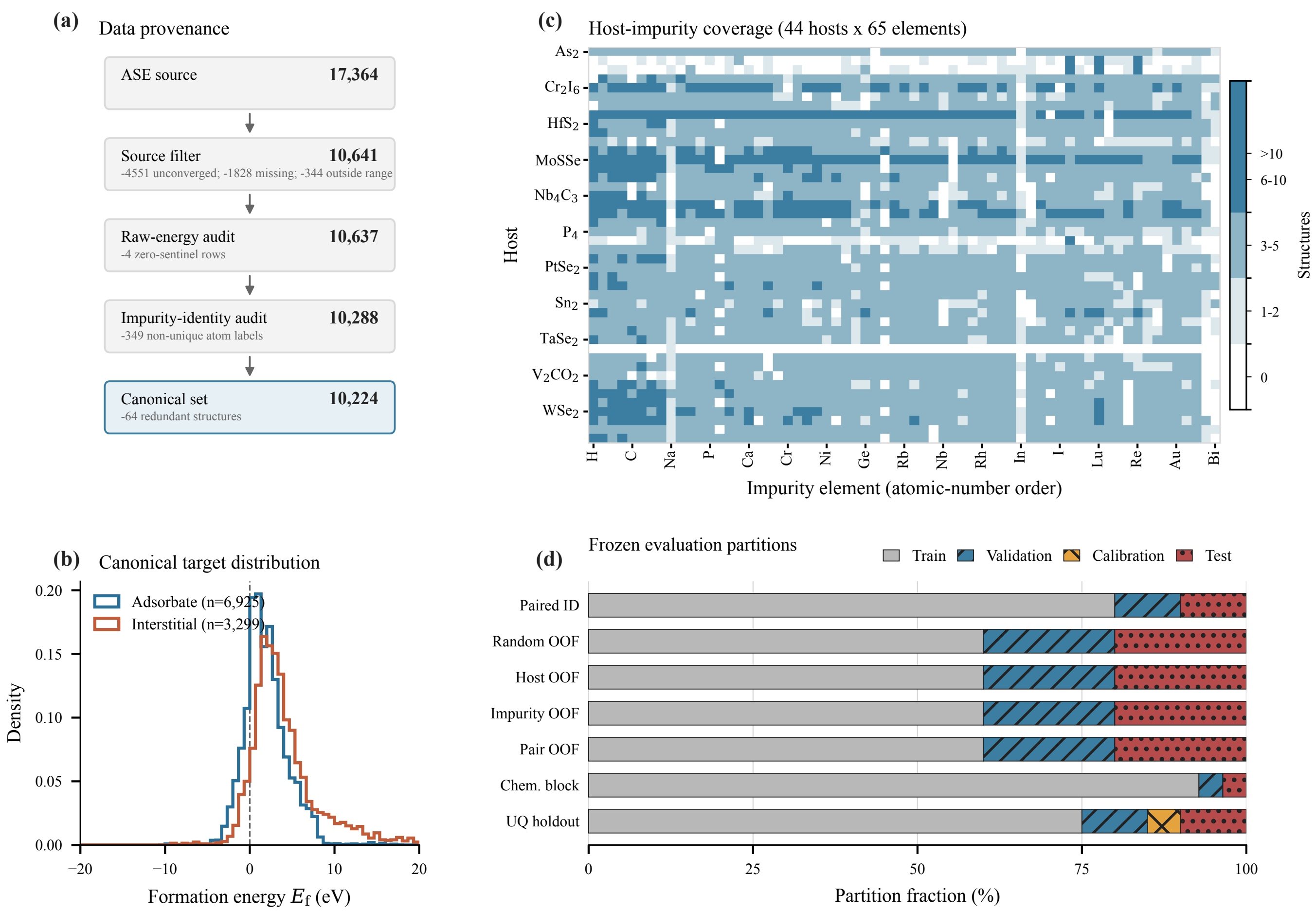

FIG. S1. Data provenance and evaluation protocol. (a) Sequential source filtering, raw-energy reconstruction audit, impurity-identity audit, and structure canonicalization. The final set excludes four rows with zero sentinels in a raw energy component, 349 rows without a unique impurity atom, and 64 redundant structures. (b) Formation-energy distributions for the two incorporation classes in the retained set. (c) Number of structures for each of 44 hosts and 65 impurity elements; white cells are unobserved combinations. (d) Train, validation, calibration, and test fractions for each evaluation regime. Bars for fivefold protocols show fold averages. All regimes cover the same 10,224 retained structures; the common 417-row exclusion set is omitted from the bars.

TABLE SI. Test MAE (eV) across the five transfer regimes. DART values are seed-averaged repaired out-of-fold predictions (fold mean); SchNet is the additive-readout baseline under the identical protocol; the descriptor column is the model selected by validation MAE within each split. Parentheses give the fold standard deviation where five folds exist.

Regime	DART	SchNet	Descriptor
Random cross-validation	0.408 (0.019)	0.571 (0.018)	0.625
Pair held out	0.442 (0.014)	0.628 (0.029)	0.674
Chemistry block	0.483	0.543	0.642
Impurity held out	0.844 (0.139)	1.692 (0.098)	1.334
Host held out	0.891 (0.145)	6.701 (2.619)	1.266

TABLE SII. Paired 2^3 architecture factorial. Values are mean MAE with 95% bootstrap intervals over five paired repeats, in eV. G denotes gated readout, E defect-environment enrichment, and P the composite pre-norm distance-gated local block. Bold marks the validation-selected variant; test MAE was not used in its selection.

Variant	G	E	P	Validation MAE	Locked test MAE
g000	0	0	0	0.422 [0.413, 0.430]	0.419 [0.410, 0.433]
g001	0	0	1	0.394 [0.375, 0.413]	0.402 [0.393, 0.413]
g010	0	1	0	0.404 [0.390, 0.417]	0.412 [0.405, 0.421]
g011	0	1	1	0.401 [0.396, 0.407]	0.410 [0.394, 0.429]
g100	1	0	0	0.401 [0.389, 0.413]	0.413 [0.407, 0.420]
g101	1	0	1	0.374 [0.357, 0.388]	0.387 [0.375, 0.400]
g110	1	1	0	0.404 [0.396, 0.410]	0.405 [0.394, 0.415]
g111	**1**	**1**	**1**	**0.371 [0.357, 0.385]**	**0.382 [0.377, 0.387]**

II. ARCHITECTURE FACTORIAL

The paired architecture experiment selected the frozen all-enabled recipe and is reported here in full so that the main text can emphasize chemical transfer and materials-physics analysis. These runs were executed under the superseded, order-sensitive graph implementation and are retained as recipe-selection provenance; no main-text claim rests on them, and every main-text number derives from the repaired representation.

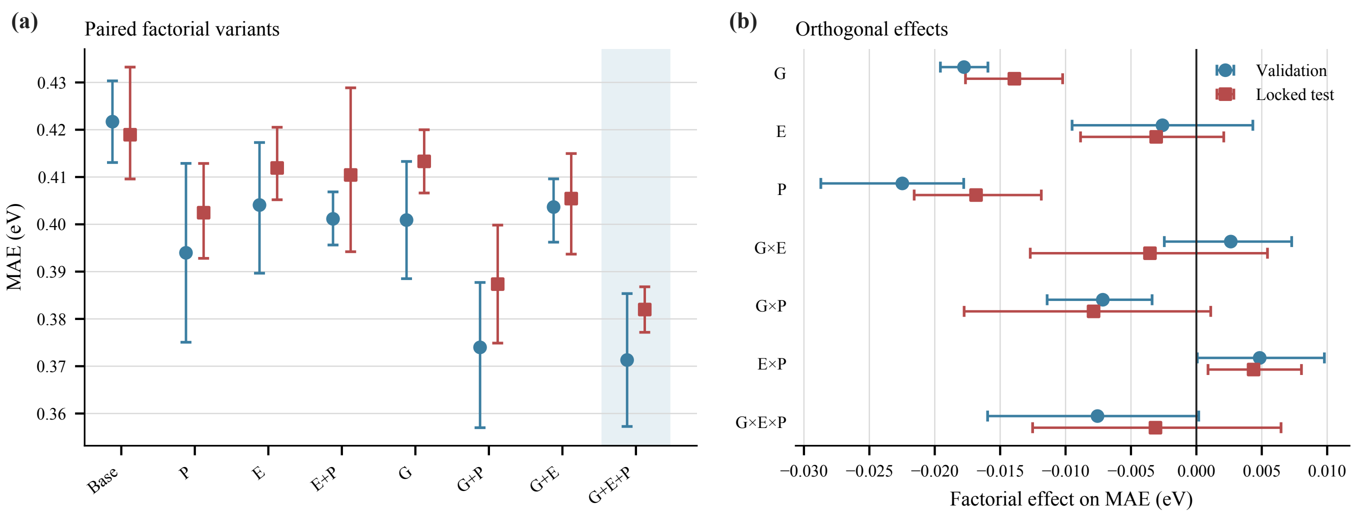

FIG. S2. Paired architecture experiment. (a) Validation and test MAE for all eight G/E/P variants; shading marks the validation-selected variant. (b) Orthogonal main effects and interactions on validation and test MAE, with 95% bootstrap intervals over paired repeats. Negative effects reduce MAE.

III. DEVELOPMENT-ERA SURVEYS AND PILOTS

A. Architecture survey

Before the formal protocol was frozen, a broad architecture survey was run on the development-era random split (the superseded "leak-free v1" partition with a 1,065-structure test set, predating the protocol-v2 sample audit and the graph repair). These values are development history: they are not comparable to the main-text protocol and support no main-text claim, but they document why the formal comparison kept SchNet as the aligned neural comparator. At matched parameter budgets, a trained equivariant MACE (0.44M parameters) reached 1.46 eV test MAE and the equivariant ViSNet (1.69M) 0.86 eV, against 0.58 eV for SchNet (0.46M), 0.54 eV for ALIGNN at 4.0M parameters, and 0.52 eV for the early defect-centered transformer (0.75M). Classical baselines spanned 1.16–2.30 eV. Separately, the MACE-MP-0 foundation model was evaluated as a zero-shot energy surrogate for the IMP2D formation-energy labels on a 100-structure sample and was found inadequate (2.62 eV, correlation $r \approx 0.04$): the database's formation-energy convention with its crystalline impurity reference is not recoverable from generic foundation-model total energies without convention-matched chemical potentials. No claim about foundation-model *relaxation* quality is made; only the label-surrogate use was tested.

TABLE SIII. Development-era survey on the superseded random split (1,065-structure test set; pre-repair implementation era). Not comparable to main-text protocol numbers.

Model	Parameters (M)	Test MAE (eV)
Mean predictor	–	2.30
Ridge regression	–	1.92
Random forest	–	1.23
LightGBM	–	1.16
MACE (small, trained)	0.44	1.46
ViSNet	1.69	0.86
SchNet	0.46	0.58
ALIGNN	4.03	0.54
Early defect-centered transformer	0.75	0.52
MACE-MP-0 (zero-shot label surrogate)	–	2.62 ($r \approx 0.04$)

B. Defect locality of the learned representation

Attribution analyses of the development-era model support the defect-centered design from inside the network. Masking single atoms and recording the prediction change (100 test structures) places $91 \pm 13\%$ of the total attribution on the impurity atom itself; the mean occlusion effect at the impurity atom exceeds that at other atoms by more than two orders of magnitude; per-atom attributions decay with distance from the defect (mean Pearson correlation 0.99 with inverse distance); and 80% of the non-defect attribution is contained within roughly 9 Å of the defect site, inside the 12 Å radial-bias horizon of the architecture. Attention shows the same concentration (200 structures): mean incoming attention to the defect atom is 32 times that to a random other atom. The learned function is therefore defect-local in the same sense as the target — the property that the main text's mean-versus-additive readout comparison probes behaviorally.

C. Held-out-host case studies

Five leave-one-host-out case studies, each withholding every structure of one named host, trace the novelty gradient concretely on the development split. Hosts with close structural relatives in training transfer at no measurable cost: MoS_2 ($n = 308$) matches the in-distribution reference exactly (0.516 versus 0.516 eV) and the Janus host MoSSe ($n = 464$) comes in below it (0.448 eV). $TaSe_2$ ($n = 222$, 1.23×) and the halide Cr_2I_6 ($n = 334$, 1.63×) degrade moderately. The hydrogenated covalent host graphane, C_2H_2 ($n = 231$, 2.40 eV), is the extreme of the distribution — and it is also the one host with no hydrogenated or purely covalent relative anywhere in the database, so the gradient anticipates the main-text attribution of held-out-host error to the structural novelty of the withheld host.

D. Cross-database and multi-source experiments

The same development era probed whether other formation-energy sources could extend or substitute IMP2D supervision; all values below carry the same superseded-split provenance and support no main-text claim. Zero-shot evaluation of an IMP2D-trained model on JARVIS vacancy data — a different defect type computed with a different DFT code — failed at the 2.3 eV (2D hosts) and 2.6 eV (3D hosts) level with negative R^2; few-shot fine-tuning onto those targets underperformed training from scratch at every budget tested (10–300 labels); and four-source joint training (IMP2D with JARVIS-2D, JARVIS-3D, and DFT-3D at tuned weights) improved the IMP2D development-split error by only about 4.5% — and leave-one-host-out replays of the same multi-source recipe degraded held-out-host transfer by 22–65% across the three hosts tested, so diluting defect-centered supervision with bulk-crystal gradients hurts exactly the transfer axis this paper measures. Together with the foundation-model surrogate test above, these results identify the cross-database barrier as one of reference conventions and defect-class mismatch rather than representation capacity, and they motivated both the frozen single-database protocol and the convention-matching requirements of the preregistered external-validation design. The retained pipeline is nevertheless two-source: the corrected JARVIS-3D pretraining (19,902 records) initializes every reported model.

E. Prospective DFT feasibility pilot

The development era also exercised, end to end, the external test the main text identifies as the most direct next step: independent first-principles calculations on candidates ranked by the model itself. From a pool of 287 generated single-impurity candidates (interstitial and substitutional mutations of database hosts), 60 were selected — the 30 lowest predicted formation energies and the 30 largest calibrated uncertainties, with a per-host diversity cap — and evaluated with plane-wave PBE calculations in Quantum ESPRESSO 7.3.1 (PSL 1.0.0 pseudopotentials, 30/180 Ry cutoffs, Γ-only, Methfessel–Paxton smearing): 125 runs in total covering the defect supercells, the 27 pristine hosts, and 38 isolated-atom chemical-potential references. A pseudopotential-library augmentation incompatibility removed the 22 La/Cs candidates and one run did not converge, leaving $N = 37$ validated formation energies.

The pilot's two findings sharpen the convention analysis above. First, the raw predicted-versus-DFT comparison is uncorrelated, and aligning the pilot's isolated-atom chemical-potential reference with the database's crystalline convention through one constant per impurity element (estimated from the paired data, as the pilot computed no bulk references) restores rank agreement — Spearman 0.35, stratified-bootstrap 95% interval $[0.04, 0.60]$, excluding zero. The cross-database barrier is therefore a per-element reference-convention offset, not a loss of rank information. Second, model ranking concentrates first-principles effort effectively: within the low-predicted-energy bucket, 70% (14/20; interval $[50\%, 90\%]$) of the converged candidates were confirmed below $+1$ eV and 40% were exothermic, against roughly 18% of the full pool predicted below the same threshold. The pilot used development-era model predictions and unrelaxed single-point energies at pilot-grade settings, so its absolute errors (median 1.7 eV on this deliberately out-of-distribution set) characterize the pilot rather than the repaired model, and no pilot number supports a main-text claim; what carries over is the demonstrated workflow — candidate generation, reference chemical potentials, pseudopotential coverage, and per-element convention alignment — which leaves convention matching as the only remaining ingredient of the preregistered external-validation design.

IV. REPRODUCIBILITY SPECIFICATION

a. Elemental feature construction. The nine elemental attribute columns of the main-text node representation are min–max normalized over atomic numbers 1–100; an unavailable van der Waals radius is replaced by the corresponding covalent radius, and the other tabulated values, including explicit unavailable or approximate entries, are fixed by the released feature table. The 128-dimensional ct-UAE table is reconstructed from the published checkpoint by applying its linear `atom_embed` layer to each one-hot elemental code, $\mathbf{U} = \mathbf{W}^{\mathsf{T}} + \mathbf{1}\mathbf{b}^{\mathsf{T}}$; after a zero-padding row is prepended, atomic number Z indexes table row Z, corresponding to row $Z - 1$ of $\mathbf{U}$.

Table SIV records the fixed model and optimization settings used in the main comparison. The resolved configuration for every run, including split and seed, is archived under `configs/prm/`; the corresponding run manifests bind those configurations to checkpoints, predictions, environments, and output hashes.

TABLE SIV. Fixed settings for the selected DART model and the predefined periodic SchNet comparator. Settings not shown as different were shared.

Setting	DART (`g111`)	Periodic SchNet (additive readout)
Atomic input	Fixed 128-dimensional ct-UAE vector, nine tabulated elemental attributes, and a learned 128-channel impurity-status embedding	Learned atomic-number embedding
Interaction stack	Three 5-Å local layers with 32 radial and 32 angular functions, followed by two four-head geometric Transformer blocks with 32 radial-bias centers over 0–12 Å and all-atom attention	Six 5-Å interaction blocks with 50 Gaussian functions
Hidden width and readout	128 channels; gated fusion of an atom-attention sum and channelwise maximum	128 channels and filters; atomwise additive scalar readout
Trainable parameters	820,558	455,937
Optimization	AdamW, batch size 64	AdamW, batch size 32
Shared schedule	150 epochs; MAE objective; learning rate 5×10^{-5} to 5×10^{-4} over a ten-epoch linear warmup, then cosine decay to 10^{-6}; weight decay 10^{-4}; gradient clipping at 5.	
Shared sampling and regularization	Training-target standardization; host-balanced sampling of 200 structures per host per epoch; 0.03-eV Gaussian target noise; no coordinate or strain augmentation. DART additionally uses 0.1 dropout.	

Factorial assignment seeds were 42–46 with paired model seeds 142–146. Transfer runs used DART seeds 242–244 and SchNet seeds 342–344, with one fixed seed per fold for random and host–impurity-pair cross-validation and three seeds per split for host, impurity, and chemistry-block evaluation. The dedicated DART uncertainty ensemble used seeds 442–446.

The descriptor search was also fixed before test aggregation. Ridge used $\alpha \in \{0.01, 0.1, 1, 10, 100\}$. Random forests used 400 trees with `max_features` in $\{0.5, 1.0\}$ and `min_samples_leaf` in $\{1, 2\}$. Histogram gradient boosting used 400 iterations, learning rate 0.05, leaf counts in $\{31, 63\}$, and L_2 regularization in $\{0, 1\}$. LightGBM used 600 trees, learning rate 0.05, leaf counts in $\{31, 63\}$, child counts in $\{20, 50\}$, L_2 regularization 1, and row/column subsampling fractions 0.9. A hyperparameter setting and then a learned family were selected independently inside each split by validation MAE; the training-target mean predictor was reported separately and never entered family selection.

V. INTERNAL UNCERTAINTY AND SELECTIVE PREDICTION

The internal ensemble analysis uses a dedicated calibration/test partition drawn from the retained IMP2D population. Its intervals therefore assess marginal calibration inside that population rather than calibration under an explicit host or impurity shift. The ensemble members were trained under the superseded graph implementation; the analysis is retained as supporting evidence for the calibration methodology, and no main-text claim rests on these values.

TABLE SV. Internal split-conformal interval performance on the dedicated test partition. These structures were excluded from ensemble fitting, checkpoint selection, and calibration, but not from the earlier architecture-development corpus. Coverage is in percent and mean width is in eV.

Nominal	Observed	Mean width	Quantile
50	52.0	0.336	0.270
80	78.3	0.752	0.604
90	87.2	1.141	0.917
95	93.7	1.847	1.484

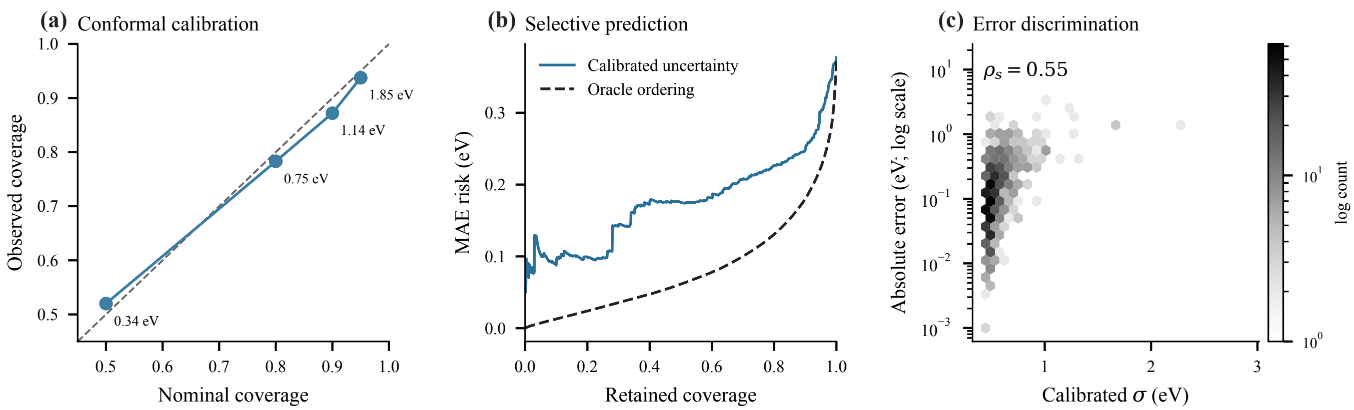

FIG. S3. Internal uncertainty evaluation on the dedicated ensemble test split. (a) Nominal versus observed split-conformal coverage; annotations give mean interval widths. (b) MAE risk as increasingly uncertain predictions are rejected, compared with oracle ordering by realized error. (c) Calibrated standard deviation versus absolute error on the dedicated test partition.

VI. RETROSPECTIVE RERANKING METRICS

These metrics compare predictions and reference energies over available DFT-relaxed IMP2D geometries, using the repaired representation's pair-held-out predictions throughout. The geometries and reference energies were produced by the same completed DFT workflow, so the analysis does not replace candidate generation or relaxation.

TABLE SVI. Retrospective out-of-fold reranking metrics from host–impurity-pair folds. Intervals are 95% nonparametric cluster-bootstrap intervals over host–impurity pairs. Accuracy gains are model minus reference in percentage points.

Metric	Model [95% interval]	Reference	Gain [95% interval]	n
Incorporation-class preference accuracy (%)	91.4 [90.0, 92.7]	66.2	25.1 [22.7, 27.6]	1730
Global exact-site accuracy (%)	68.9 [66.7, 71.1]	23.4	45.5 [43.4, 47.7]	1730
Global reranking regret (eV)	0.142 [0.116, 0.172]	–	–	1730
Within-class exact-site accuracy (%)	73.6 [72.0, 75.1]	37.1	36.5 [35.0, 38.0]	3140
Within-class top-2 accuracy (%)	91.5 [90.3, 92.7]	62.1	29.4 [28.2, 30.6]	2145
Within-class reranking regret (eV)	0.099 [0.083, 0.118]	–	–	3140

VII. DESCRIPTIVE ERROR HETEROGENEITY

Across host and impurity groups, the Spearman associations between group sample count and group MAE were -0.523 and -0.001, respectively. The largest group MAEs occurred for host Bi2I6 (1.851 eV; $n = 52$) and impurity In (1.502 eV; $n = 35$). Because group size covaries with chemistry, these are descriptive associations rather than evidence that sample count alone causes the observed errors.

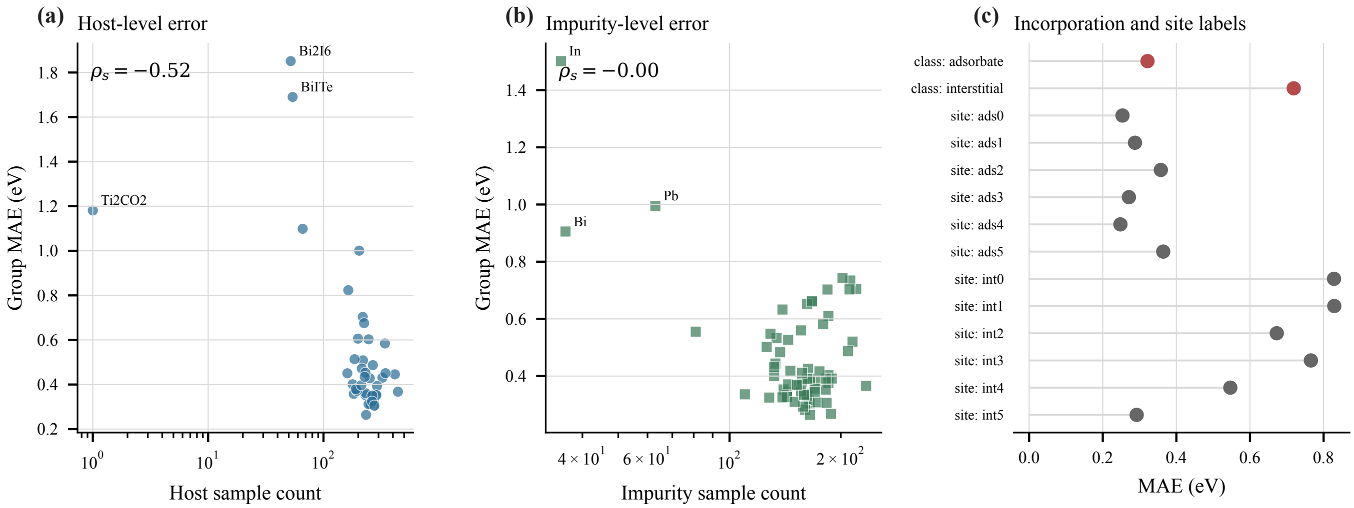

FIG. S4. Error heterogeneity from DART pair-held-out out-of-fold predictions. (a,b) Host- and impurity-group MAE versus the number of retained structures; labels mark the three largest group MAEs and ρ_s is the descriptive Spearman association. (c) MAE by incorporation class and source site label. These summaries were not used for architecture, hyperparameter, or model selection.

Figure S5 shows the parity of the repaired pair-held-out out-of-fold predictions by incorporation class; the class MAEs match the main-text values, and the interstitial cloud carries both the larger spread and the extreme-target tail. Figure S6 reports the canonical family-level macro errors with identity and structure counts, alongside the out-of-plane expansion-factor diagnostic behind the XF ≤ 2 gate. Figure S7 maps the descriptive per-impurity-element errors and retained counts. Figure S8 renders the four highest-error non-pathological interstitials and their matched bottom-quartile adsorbate controls, aligned column by column by the frozen matching rule (the first-ranked interstitial has no eligible control); the extreme interstitial errors concentrate at extreme formation-energy targets ($E_f \approx 12$–$19\,\mathrm{eV}$ for three of the four). The case panel is illustrative (exploratory tier under the frozen evidence rules) and supports no gated main-text claim.

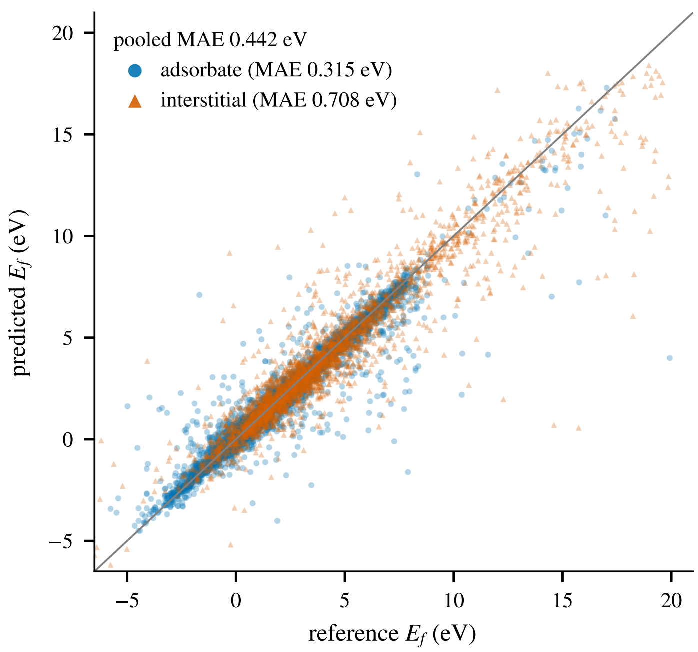

FIG. S5. Parity of repaired pair-held-out out-of-fold predictions for the 10,224 retained structures, by incorporation class. Marker shape and color encode the class; the inset reports pooled and per-class MAE.

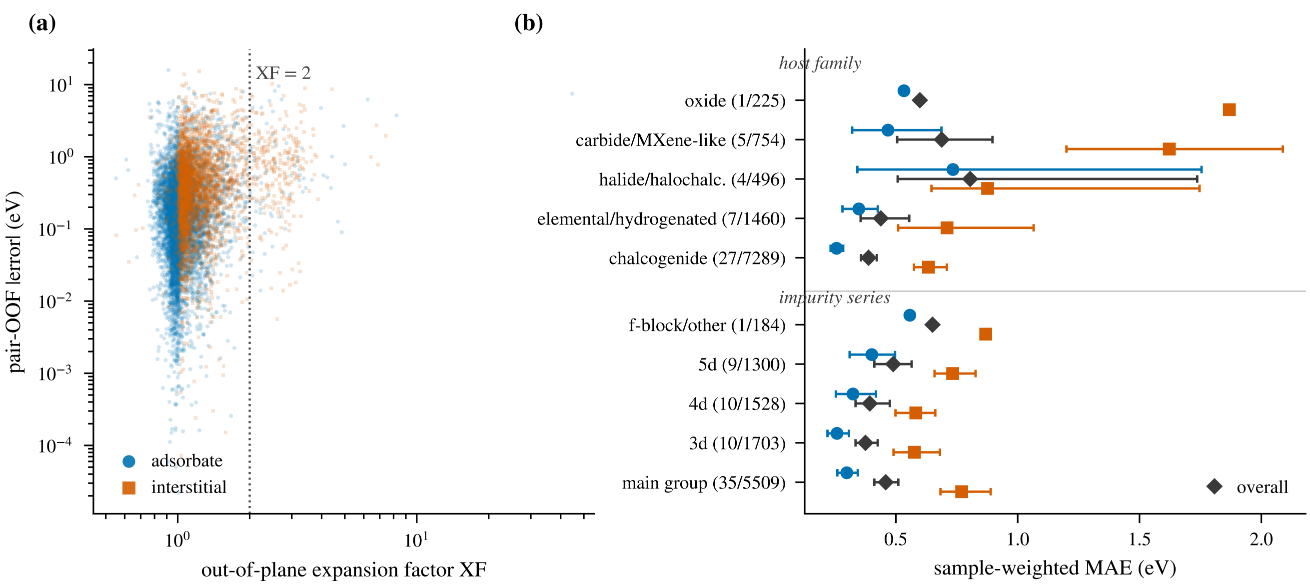

FIG. S6. Macro panels of the canonical error map. (a) Pair-held-out absolute error against the database out-of-plane expansion factor; the dotted line marks XF = 2, beyond which structures are treated as reconstructed and analyzed separately. (b) Host-family and impurity-series sample-weighted macro MAEs with pair-clustered intervals; group labels give (identities/structures), and the diamond marks the class-pooled value.

FIG. S7. Per-impurity-element MAE of the repaired pair-held-out out-of-fold predictions. Each occupied cell reports the element and its retained structure count; gray cells are elements outside the 65-impurity set. The color scale is clipped at the 95th percentile of the per-element MAEs. These are descriptive summaries of the same out-of-fold predictions; no per-element claim is gated.

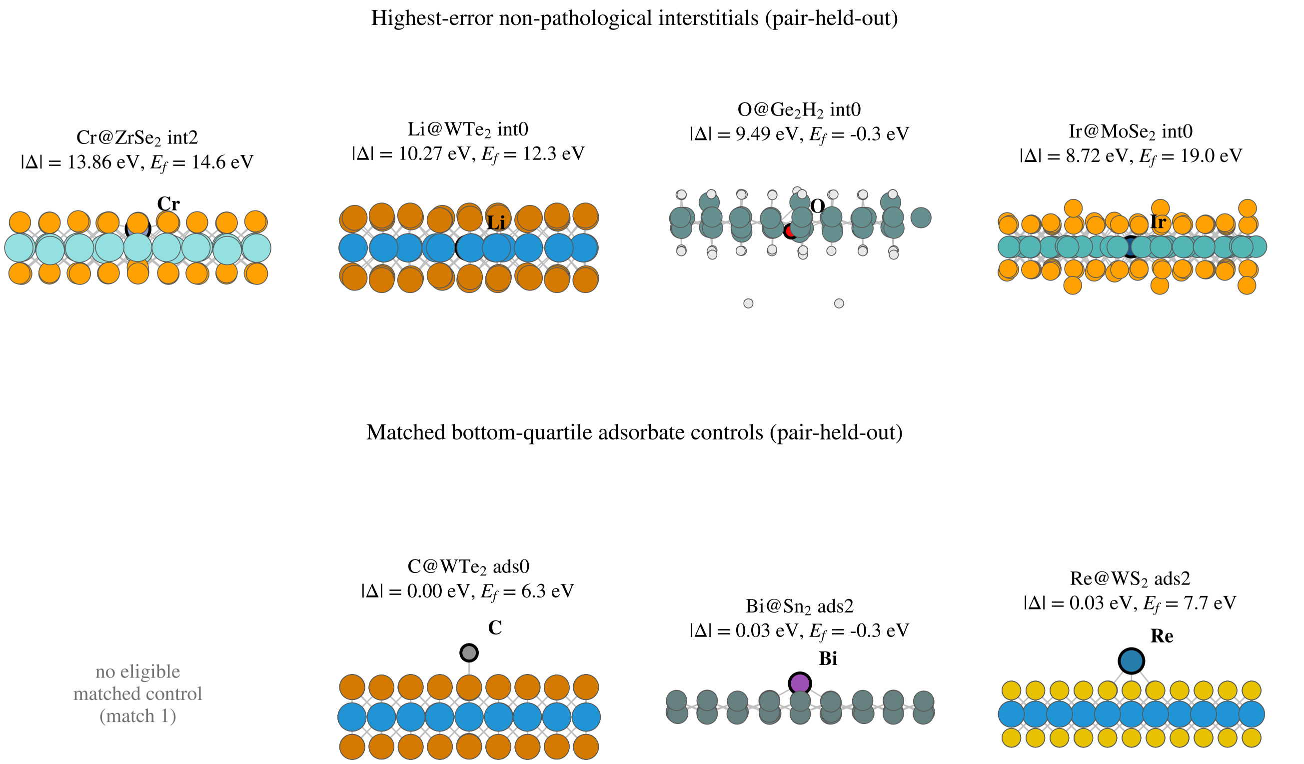

FIG. S8. Structures selected by the frozen case rule, rendered from the source-database geometries (side view, periodic images tiled in plane, impurity atom outlined in black and labeled). Top row: the four highest-error non-pathological interstitials under pair holdout, in match order. Bottom row: the matched bottom-quartile adsorbate controls under the same frozen rule. Panel headers give the pair-held-out absolute error and the reference formation energy. Illustrative only; no gated claim.

VIII. POST-HOC SCHNET GRAPH-READOUT SENSITIVITY

The predefined periodic SchNet comparator follows the original energy-oriented construction [1]: its learned atom-wise scalar contributions are summed to produce the graph prediction. The present defect formation energy is non-extensive with respect to host atom count, and prior atomistic studies show that graph pooling can materially affect intensive or localized quantum-property prediction [2]. Because the target may still contain long-range host response, this comparison tests the specific SchNet readout rather than assuming strict spatial locality. We therefore performed one bounded post-hoc sensitivity analysis after completing the main comparison.

The intervention changed only the graph readout from atomwise addition to mean pooling. It retained all five predefined host-held-out folds, seeds 342–344, the SchNet interaction stack, learned atomic-number input, optimizer, learning rate schedule, host-balanced sampler, target normalization, target-noise stream, gradient clipping, and 150-epoch budget. Predictions were averaged over the three seeds within each fold. The pooled mean-minus-add difference in absolute error was evaluated with 20,000 bootstrap samples clustered by the 44 host materials.

TABLE SVII. Post-hoc SchNet graph-readout sensitivity under the fixed host-held-out protocol. Each fold prediction averages seeds 342–344. The difference is mean-readout MAE minus additive-readout MAE, in eV. The pooled host-cluster bootstrap 95% interval for this difference is [-7.704, -1.256] eV. The prespecified additive-readout result remains the main comparator.

Partition	n	Add MAE	Mean MAE	Mean − add
Fold 1	2023	3.882	2.179	-1.704
Fold 2	2019	8.199	2.068	-6.132
Fold 3	2069	5.484	2.530	-2.954
Fold 4	2057	5.466	2.468	-2.998
Fold 5	2056	10.474	3.501	-6.973
Pooled OOF	10224	**6.703**	**2.552**	**-4.151**

Mean pooling reduced the host-held-out MAE in all 5/5 folds. Across individual samples, the Spearman association between atom count and absolute error changed from 0.278 for additive pooling to 0.010 for mean pooling. Across the 44 hosts, the corresponding association between median atom count and host MAE changed from 0.312 to 0.218. These descriptive associations are consistent with a size-sensitive additive readout, but atom count covaries with host chemistry and therefore does not identify a standalone causal mechanism. The intervention establishes that readout choice contributes to the extreme additive-SchNet error; it neither replaces the predefined main comparator nor attributes the remaining DART–SchNet difference to a single component.

[1] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, SchNet – a deep learning architecture for molecules and materials, The Journal of Chemical Physics **148**, 241722 (2018).

[2] D. Buterez, J. P. Janet, S. J. Kiddle, D. Oglic, and P. Liò, Modelling local and general quantum mechanical properties with attention-based pooling, Communications Chemistry **6**, 262 (2023).